Learned Modulation of SE(3) Dynamical System Policies via Gaussian Process Regression

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***Abstract*— While existing motion policies effectively generalize within the patterns of a training set, it is unrealistic to expect them to extrapolate to entirely novel scenarios that fall outside the demonstrated distribution. For instance, a policy trained for reaching cannot inherently account for unforeseen environmental obstacles or internal kinematic constraints at runtime unless explicitly instructed otherwise. To bridge this gap, a robust modulation strategy is essential to adapt nominal behaviors to these real-time challenges without requiring a complete retraining of the underlying policy. Although modulation for translational dynamical systems (DS)-based motion policy is well-studied, orientation modulation yet remains overlooked, despite its necessity for expressive and adaptive robotic behavior. Thus, we propose a framework for learning to modulate orientation motion policies using corrections derived from either offline demonstrations or online generation. Leveraging the techniques from Riemannian geometry, we represent the desired modulation by a compact vector comprising of scaling factor and an axis-angle rotation within the tangent space of manifold. We also employ Gaussian Process (GP) regression to learn the mapping from orientation states to these parameters, ensuring the modulation preserves the stability certificates of the nominal dynamics. We extensively validate our approach in both simulation and real robot experiments, across various applications from obstacle avoidance to manipulability optimization.**

I. Introduction

Dynamical System (DS)-based motion policies encode learned trajectories as globally stable autonomous vector fields, enabling reactive real-time execution with Lyapunov-certified convergence [1]–[3]. The Linear Parameter-Varying formulation (LPV-DS) significantly extended this capability by utilizing state-dependent gain mixing to capture complex, curved trajectories [2]. However, despite their rigorous stability guarantees, these learned DS policies are inherently rigid. Once a model is trained, its vector field cannot be easily reshaped without retraining from scratch. In practice, a natural requirement is the ability to modify a learned DS dynamically to satisfy execution-time objectives—such as avoiding kinematic singularities, enforcing task-specific constraints, or adapting to new environments—without discarding the original demonstrated structure. Addressing these runtime challenges calls for a principled modulation framework: a systematic method to represent, learn, and apply state-dependent corrections to a nominal DS output. While modulation strategies for translational dynamical systems are well-studied—such as the Locally Modulated Dynamical Systems (LMDS) framework [4]—orientation modulation remains largely overlooked. Because expressive and adaptive robotic behavior fundamentally occurs in $SE(3)$, extending modulation to rotational motion is critical. However, the non-Euclidean nature of orientation spaces, such as the unit quaternion manifold S^3, makes this non-trivial and requires careful Riemannian geometric formulations to

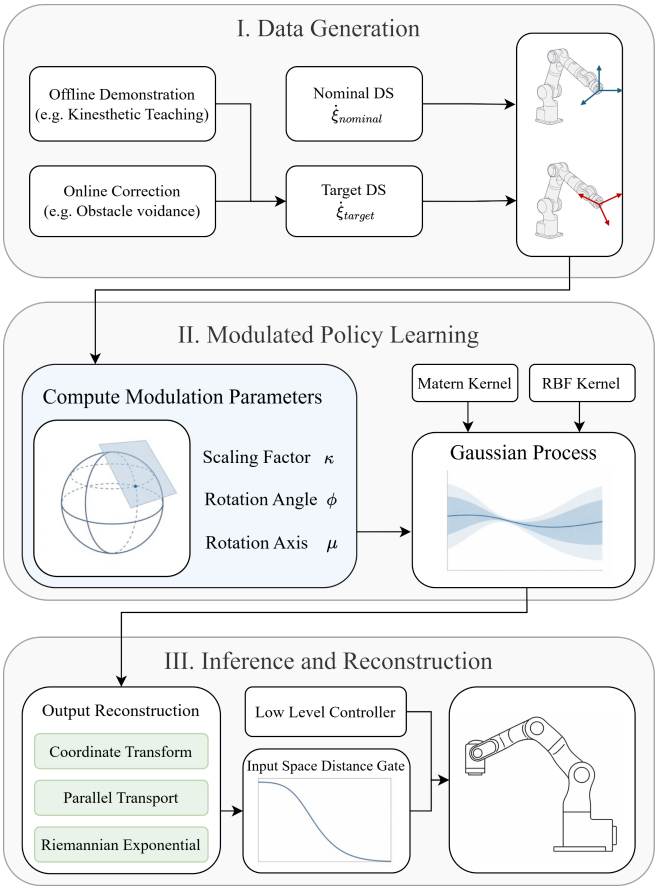

Fig. 1: Framework for Learning SE(3) Orientation Modulation with Stability and Safety Certificates. The pipeline operates in three stages. **(I) Data Generation:** Modulation training data are obtained from either kinesthetic demonstrations or an optimization oracle (e.g., manipulability maximization). The nominal DS output and the target DS output are projected into the attractor-aligned tangent space to compute their difference. **(II) Learning Modulation Policies:** A Gaussian Process with a Matérn κ=2.5 kernel is trained to map the current orientation state—encoded via the Riemannian logarithmic map—to compact modulation parameters $(\kappa, \phi\mu)$ representing speed scaling and axis-angle rotation. A weighted merge with an RBF distance gate preserves the Lyapunov stability certificates of the nominal DS. **(III) Reconstruction and Execution:** The modulated tangent-space vector is reintegrated onto the manifold via the exponential map, yielding a corrected velocity command sent to the 7-DoF Franka Panda robot.

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maintain stability and correctness [5]–[7]. To bridge this gap, this work proposes a novel framework for learning to modulate orientation DS policies on S^3. Building upon the foundations of Quaternion-DS [8], our key insight is that any desired correction to the nominal orientation output can be expressed as a compact 4-dimensional vector within the attractor-aligned quaternion tangent space. This vector comprises a scaling factor κ and an axis-angle rotation $\phi\boldsymbol{\mu}$. By leveraging Riemannian exponential and logarithmic maps alongside parallel transport, we extract these geometric modulation parameters from an oracle or offline demonstrations. We subsequently employ Gaussian Process (GP) regression to learn the mapping from orientation states to these extracted parameters. During inference, the GP is queried in closed form using the current state expressed as the Riemannian logarithm at the policy attractor. To ensure safe execution, an RBF distance gate suppresses modulation in undemonstrated regions, smoothly fading out the learned correction so the system safely falls back to the convergence-guaranteed nominal DS. The proposed framework is agnostic to the source of the oracle. As a concrete validation, we apply it to a manipulability-aware adaptation task on a 7-DoF Franka Emika Panda robot. In this scenario, the oracle is a reference DS pre-modulated to steer the end-effector away from kinematic singularities, and the GP successfully learns to replicate this behavior seamlessly from the state alone. The core contributions of this work are:

1) A geometric decomposition of orientation DS modulation into a compact 4D parameter space (speed-scaling and axis-angle rotation) within the attractor-aligned quaternion tangent frame.
2) A robust GP training and inference pipeline that utilizes a Riemannian distance-based gate to suppress unverified modulations, preserving the stability certificates of the nominal dynamics.
3) Extensive validation in both simulation and real-robot experiments on a manipulability-aware pouring task, demonstrating that the framework reproduces corrections with high fidelity.

The remainder of this paper is organized as follows: In Section II we summarize related work on modulation of existing policy. Section III introduces the mathematical background on Gaussian Process, Riemannian geometry and the nominal DS formulation. Section IV details our proposed modulation framework, including the stability analysis. Section V presents our experimental setup and results for both simulation and real-world robot tasks.

II. Related Work

a) DS-based motion policies.: Dynamical Systems (DS) have become a cornerstone of reactive robot control due to their ability to provide global asymptotic stability (GAS) via Lyapunov theory. Early approaches, such as the Stable Estimator of Dynamical Systems (SEDS) [1], utilized Gaussian Mixture Models (GMM) constrained by Lyapunov stability to learn motions from human demonstrations. This was later extended by the Linear Parameter-Varying (LPV-DS) framework [9], which offers greater flexibility in capturing complex, non-linear trajectories by mixing linear systems while maintaining rigorous stability certificates.While these methods primarily focus on Euclidean spaces $\mathbb{R}^n$, recent research has moved toward learning policies on non-Euclidean manifolds. Formulations such as Quaternion-DS [10] and Riemannian-based approaches [11] ensure that orientation trajectories remain on the unit sphere S^3, avoiding the singularities and discontinuities associated with Euler angles. Our work builds upon these stable $SE(3)$ formulations, utilizing them as a nominal behavior layer.

b) Policy Modulation and Correction: Even a perfectly learned DS policy may require runtime adaptation to handle environmental changes or kinematic constraints. Policy modulation addresses this by applying state-dependent transformations to the nominal vector field. The Locally Modulated Dynamical Systems (LMDS) framework [12] introduced a method to reshape vector fields for obstacle avoidance by using a modulation matrix that preserves the original equilibrium point.However, correcting orientation policies is significantly more complex due to the non-Euclidean geometry of $SO(3)$. Standard additive corrections often violate unit-norm constraints, necessitating geometric transformations within the tangent space [13]. We extend these concepts by proposing a modulation strategy that operates within the attractor-aligned tangent space of S^3, allowing for intuitive corrections—such as scaling and rotation—that respect the underlying manifold.

c) GP-based motion modulation.: Gaussian Processes (GPs) provide a principled, uncertainty-aware mechanism for learning motion modulations. A foundational work by Kronander et al. [4] proposed using GPs to incrementally learn a modulation matrix from corrective human demonstrations, ensuring the reshaped DS remains stable. This approach allows the robot to adapt its behavior locally in regions where the original policy is insufficient.In our framework, we specifically leverage the predictive variance of the GP—a key advantage highlighted in recent residual learning works [14]—to implement a "distance-based gate." This ensures that the modulation only activates in high-confidence regions; in unexplored states, the system gracefully reverts to the nominal stable DS. Unlike the additive corrections found in [15], our GP regresses a compact 4D geometric parameter set $(\kappa, \phi, \boldsymbol{\mu})$ to perform scaling and rotation in the tangent space, providing a more structured and geometrically consistent adaptation for $SE(3)$ tasks.

III. Background

A. Quaternion Dynamical System

We model end-effector motion as first-order autonomous dynamical systems on $\mathrm{SE}(3)$. The robot state is $\xi = (p, q) \in \mathbb{R}^3 \times \mathbb{S}^3$, where $p \in \mathbb{R}^3$ is the Cartesian position and $q \in \mathbb{S}^3$ is the unit quaternion. Position and orientation dynamics are decoupled and learned independently. The translational motion is learned with the classic LPV-DS framework [9],

whereas the rotational motion is learned with a Quaternion-DS formulation [8]. As our contribution focuses on the orientation domain, we provide a brief preliminary on Quaternion-DS below and direct readers to [9] for a comprehensive review of classic LPV-DS.

The quaternion dynamical system, or Quaternion-DS, is formulated on the tangent plane of the quaternion space. For clarity, we denote elements on tangent plane in fraktur typeface; i.e., $\mathfrak{q} \in T\mathcal{M}$. Quaternion-DS is a discrete system that generates the next desired orientation as follows:

$$\hat{\mathfrak{q}}_{att} = \sum_{k=1}^{K} \gamma_k(q;\, \Theta_\gamma) A_k \log_{q_{att}}(q), \tag{1}$$

where q is current quaternion, q_{att} is attractor, $\hat{\mathfrak{q}}_{att}$ is the predicted next orientation on the tangent plane at the attractor, and K is the number of linear time invariant (LTI) systems. The Riemannian logarithmic map $\log_{q_{att}}(q)$ computes the tangent vector representing the deviation between the current orientation q and the target q_{att} [16]–[18]. The mixing function $\gamma_k(\cdot)$ quantifies the weight of each LTI system, whose parameters Θ_γ are estimated via a Gaussian mixture model fit to the quaternion trajectory on the tangent plane at q_{att}. Each LTI system, parameterized by A_k, is then learned by solving a semi-definite program detailed in [8].

B. Riemannian Geometry on Unit Sphere

A Riemannian manifold is a smooth manifold $\mathcal{M}$ equipped with a smooth, symmetric, and positive-definite metric tensor, known as g. It acts as a smoothly varying inner product on tangent spaces $T\mathcal{M}$, allowing for the measurement of distances, angles, and other geometric properties. Common examples of Riemannian manifolds include standard Euclidean space $\mathbb{R}^n$, hyperbolic space $\mathbb{H}^n$, and the 3D rotation group $SO(3)$. Hereafter, we focus specifically on the unit sphere and introduce the following fundamental operations pertaining to this particular manifold.

The exponential map in Riemannian geometry, denoted by $\exp_p : T_p\mathcal{M} \to \mathcal{M}$, projects a tangent vector $\mathfrak{v} \in T_q S^3$ onto the manifold at a base point $u \in \mathbb{S}^N$ along a geodesic:

$$\mathrm{Exp}_u(\mathfrak{v}) = \cos(\|\mathfrak{v}\|)\, u + \sin(\|\mathfrak{v}\|)\, \frac{\mathfrak{v}}{\|\mathfrak{v}\|}. \tag{2}$$

The logarithmic map, denoted by $\log_u : \mathcal{M} \to T_u\mathcal{M}$, is the inverse of the exponential map, which projects a point $v \in \mathbb{S}^N$ onto the tangent space at a base point u:

$$\mathrm{Log}_u(v) = d(u,v)\, \frac{v - u^T v u}{\|v - u^T v u|}. \tag{3}$$

Note that this yields the shortest tangent vector at u whose geodesic reaches v. The norm of this vector represents the geodesic distance $d(u,v)$ between the points, such that $\|\log_u(v)\| = d(u,v)$.

The parallel transport, denoted by $\Gamma_{u\to v} : T_u\mathcal{M} \to T_v\mathcal{M}$, maps $\mathfrak{v}$ from u to v along the connecting geodesic:

$$\Gamma_{u\to v}(\mathfrak{v}) = \mathfrak{v} - \frac{\log_u(v)^T \mathfrak{v}}{d(u,v)^2} \left(\log_u(v) + \log_v(u)\right) \tag{4}$$

which preserves the inner product and norm of v under transport.

C. Locally Modulated Dynamical Systems (LMDS)

The Locally Modulated Dynamical Systems (LMDS) framework, introduced in [4], provides a methodology for reshaping the trajectories of an existing, time-invariant autonomous dynamical system while preserving its stability. Let an original, unperturbed dynamical system be defined by a continuous function $f : \mathbb{R}^N \mapsto \mathbb{R}^N : \dot{x} = f(x)$, where $x \in \mathbb{R}^N$ represents the state variable (e.g., a Cartesian position vector). The LMDS framework modifies this original motion plan by introducing a state-dependent modulation matrix $M(x) \in \mathbb{R}^{N\times N}$. The reshaped dynamics are governed by:

$$\dot{x} = g(x) = M(x) f(x). \tag{5}$$

While $M(x)$ can take various forms, a particularly robust parameterization involves state-dependent rotation and speed-scaling. The specific modulation function is defined as:

$$M(x) = (1 + \kappa(x)) R(x), \tag{6}$$

where $R(x)$ is a state-dependent rotation matrix parameterized by rotational angle $\phi(x)$ and rotational axis $\mu(x)$. $\kappa(x) > -1$ is a state-dependent scalar function controlling the speed of the motion ($\kappa(x) = 0$ retains the original speed). To learn these state-dependent modulation parameters from a set of training demonstrations, we apply Gaussian Process Regression (GPR), which is detailed next.

D. Gaussian Process Regression

A Gaussian Process (GP) defines a distribution over functions $f : \mathcal{X} \to \mathbb{R}$ fully specified by a mean function $m(x)$ and a covariance (kernel) function $k(x, x')$:

$$f(x) \sim \mathcal{GP}(m(x),\ k(x,x'))\,. \tag{7}$$

A zero-mean prior, $m(x) = 0$, is usually adopted and deemed appropriate when outputs represent deviations from a nominal behavior. Given N noisy observations $y = [y_1, \ldots, y_N]^\top$ at inputs $X = \{x_i\}_{i=1}^N$ with noise variance σ_n^2, the predictive distribution at a test point x^* is Gaussian with mean and variance as follows:

$$\bar{f}^* = k_*^\top \left(K + \sigma_n^2 I\right)^{-1} y, \tag{8}$$

$$\mathrm{Var}[f^*] = k(x^*, x^*) - k_*^\top \left(K + \sigma_n^2 I\right)^{-1} k_*, \tag{9}$$

where K is the $N \times N$ kernel matrix with entries $K_{ij} = k(x_i, x_j)$, $k_* = [k(x^*, x_1), \ldots, k(x^*, x_N)]^\top$ is the vector of covariances between the query point and the training data, and I is the $N \times N$ identity matrix. The choice of the covariance function $k(\cdot,\cdot)$ governs the smoothness and generalization properties of the inferred model.

IV. Modulated Quaternion Dynamical Systems

The proposed framework follows a two-stage pipeline. In the *training* phase, we extract geometric modulation parameters from the modulation data and train a GP. In the *inference* phase, the learned model is queried to reshape the nominal DS velocity while maintaining stability guarantees.

A. Geometric Parameter Extraction and GP Training

We first define the modulation training set as M pairs of unit quaternions $\{q_{\text{in}}^i, \hat{q}_{\text{out}}^i\}_{i=1}^M$, where q_{in}^i denotes the current orientation and $\hat{q}_{\text{out}}^i$ is the desired target orie ntation under modulation. Given a nominal DS $f(\cdot)$, our objective is to learn a modulated DS $\hat{f}(\cdot)$, such that the modulated response aligns with the behavior specified in the training set: $\hat{f}(q_{\text{in}}^i) \approx \hat{q}_{\text{out}}^i$. We first compute the original output denoted by $q_{\text{out}}^i := f(q_{\text{in}}^i)$. These are computed by evaluating the nominal DS at all inputs $q_{\text{in}}^i,\ i = 1, \ldots, M$ in the modulation training set. To extract the geometric parameters θ_i corresponding to each pair $\{q_{\text{out}}^i, \hat{q}_{\text{out}}^i\}$, we first need to transform each unit quaternion into the appropriate form by recognizing the Riemannian structure of $\mathbb{S}^3$ and leveraging the operators outlined in Sec. III-B. We begin by projecting each output quaternion onto the tangent plane of its corresponding input via the logarithmic map defined in Eq. 3:

$$\mathfrak{q}_{\text{in}}^{\text{out}} = \log_{q_{\text{in}}}(q_{\text{out}}); \quad \hat{\mathfrak{q}}_{\text{in}}^{\text{out}} = \log_{q_{\text{in}}}(\hat{q}_{\text{out}}), \tag{10}$$

where we omit the index i for notational simplicity hereinafter. We then apply parallel transport as in Eq. 4 to group each resulting output into a common tangent plane defined by the attractor q_{att}:

$$\mathfrak{q}_{\text{att}}^{\text{out}} = \Gamma_{q_{\text{in}} \to q_{\text{att}}}(\mathfrak{q}_{\text{in}}^{\text{out}}); \quad \hat{\mathfrak{q}}_{\text{att}}^{\text{out}} = \Gamma_{q_{\text{in}} \to q_{\text{att}}}(\hat{\mathfrak{q}}_{\text{in}}^{\text{out}}). \tag{11}$$

This tangent space is a 3D hyperplane embedded in $\mathbb{R}^4$, for which we construct an orthonormal basis $\mathbf{B} \in \mathbb{R}^{4\times 3}$ spanning the null space of q_{att}:

$$\mathbf{B} = [b_1, b_2, b_3], \quad \text{s.t. } b_i^\top q_{\text{att}} = 0, \quad \forall i \in 1, 2, 3. \tag{12}$$

We then apply the coordinate transformation by projecting the tangent vectors onto the basis $\mathbf{B}$ using a least-squares approach:

$$u = \mathbf{B}^\dagger \mathfrak{q}_{\text{att}}^{\text{out}}, \quad \hat{u} = \mathbf{B}^\dagger \hat{\mathfrak{q}}_{\text{att}}^{\text{out}}, \tag{13}$$

where $\mathbf{B}^\dagger$ denotes the Moore-Penrose pseudoinverse, yielding the 3D reduced representations $u, \hat{u} \in \mathbb{R}^3$. We can then extract the geometric parameters $\theta_i = [\kappa, \mu, \phi]$ from the pairs $\{u, \hat{u}\}$ as follows:

$$\kappa = \frac{|\hat{u}|}{|u|} - 1, \quad \phi = \arccos\left(\frac{u^\top \hat{u}}{|u||\hat{u}|}\right), \quad \boldsymbol{\mu} = \frac{u \times \hat{u}}{|u \times \hat{u}|}. \tag{14}$$

A Gaussian Process (GP) regression model is employed to learn the mapping from the orientation state to the extracted modulation parameters θ_i:

$$\theta(x) \sim \mathcal{GP}\left(m(x), k(x, x')\right), \tag{15}$$

where the mean function $m(x)$ is assumed to be zero, representing a lack of prior bias, while the covariance function $k(x, x')$ encodes the smoothness and correlation between different inputs. The input features x_i are defined as the logarithm map of the current orientation w.r.t. the attractor, i.e., $\log_{q_{\text{att}}}(q_{\text{in}}^i)$, thus respecting the Euclidean nature of the regression model. Since the modulation parameters θ is multidimensional, we employ independent GPs for each dimension while sharing hyperparameters to maintain computational efficiency.

B. Inference and Modulation at Runtime

At each control step t, the nominal DS provides a desired orientation q_{out}^t. To apply the learned modulation, we first acquire the reduced representation u_t, following the steps from Eq. 10 to Eq. 13. We then query the learned GP regression model using the feature input $x_t = \log_{q_{\text{att}}}(q_{\text{in}}^t)$ to retrieve the predicted modulation parameters $\hat{\theta} = [\hat{\kappa}, \hat{\mu}, \hat{\phi}]$ as in Eq. 8. From these regressed parameters, we construct the modulation matrix M_t as defined in Eq. 6 and multiply it by u_t to obtain the modulated reduced representation $\hat{u}_t$. To recover the unit quaternion representation, we need to perform a series of reverse operations. The complete pipeline, outlined in Algorithm 1, constitutes the modulated dynamical system $\hat{f}(\cdot)$.

Algorithm 1 Computing modulated target orientation

Require: Current orientation q_{in}^t, attractor q_{att}, trained $\mathcal{GP}$
Ensure: Modulated target orientation $\hat{q}^{\text{out}}$

```
for each time step t do
    Compute input feature: x_t = log_{q_att}(q_in^t)
    Compute nominal output: q_out^t = f(q_in^t)
    Apply Eqs. 10–13 to get u_t
    Apply GPR: (θ̂_t, σ̂_t) ← GP(x_t)
    Construct modulation matrix: M_t = (1+κ̂_t) R(φ̂_t μ̂_t)
    Apply modulation: û_t = M_t u_t
    Transform coordinates: q̂_att^out = B û_t
    Parallel transport: q̂_in^out = Γ_{q_att→q_in^t}(q̂_att^out)
    Output retrieval: q̂^out = exp_{q_in^t}(q̂_in^out)
end for
return Modulated orientation q̂^out
```

C. Distance-Based Uncertainty Gating

In unfamiliar or unexplored regions, data-driven models like Gaussian Processes may extrapolate unpredictably. To mitigate this risk, we introduce a spatial RBF distance gate, $g(x_t)$, which smoothly attenuates the predicted modulation parameters $\hat{\mathbf{y}}_t = [\hat{\kappa}_t, \hat{\phi}_t \hat{\boldsymbol{\mu}}_t^\top]^\top$ during inference:

$$\hat{\mathbf{y}}_t^{\text{gated}} = g(x_t) \cdot \hat{\mathbf{y}}_t,$$
$$\text{where} \quad g(x_t) = \exp\left(-\frac{\min_{x_i \in \mathcal{X}_{\text{train}}} \|x_t - x_i\|^2}{2\sigma_g^2}\right). \tag{16}$$

Here, $\mathcal{X}_{\text{train}}$ represents the set of states encountered during training, and the hyperparameter σ_g controls the sensitivity of the decay. As the robot traverses away from the training distribution, $g(x_t) \to 0$. This spatial attenuation forces the scaling factor κ and rotation angle ϕ to zero, causing the modulation matrix $M(x_t)$ to smoothly approach the identity matrix I. Consequently, the system gracefully degrades to the safe, nominal vector field, ensuring that stability certificates are never violated in uncertain states.

D. Stability Analysis

The stability of the modulated system can be analyzed from two perspectives. First, the global asymptotic stability (GAS) of the nominal orientation DS was established in the

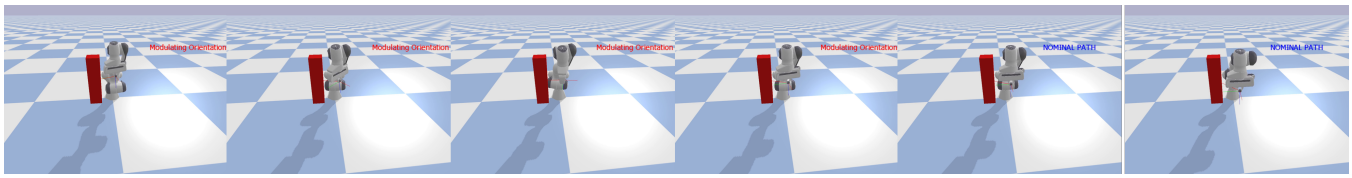

Fig. 2: Execution of the obstacle avoidance task. The robot approaches a predefined spatial obstacle (red block) that lies on its baseline trajectory. The learned GP model activates the necessary geometric corrections, through the learned policy ("Modulating Orientation") to smoothly steer the end-effector around the object. Once the obstacle is cleared, the RBF distance gate suppresses the modulation, allowing the system to safely return to the "NOMINAL PATH".

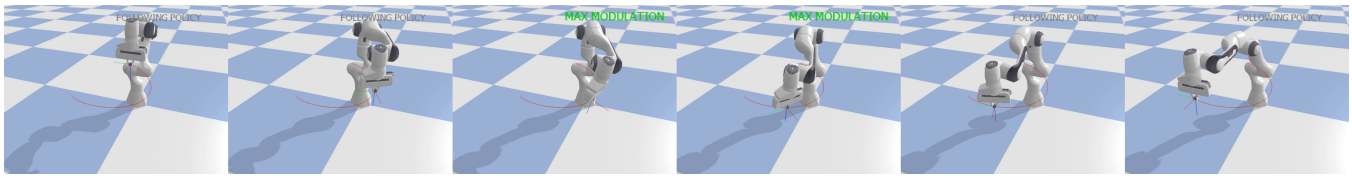

Fig. 3: Execution of the online manipulability maximization task. The Franka Panda robot begins by following the nominal dynamical system policy. As it enters a region with lower kinematic manipulability, the framework dynamically generates target orientations to improve the arm's posture (indicated by "MAX MODULATION"). The GP seamlessly applies these geometric corrections to steer the joints away from singularities before smoothly reverting back to the nominal policy.

work [8]. The GAS was proved in the tangent space of the attractor $T_{q_{\text{att}}}\mathcal{M}$: if the system evolves in this tangent space and converges to the origin asymptotically, then the actual system converges to the true attractor asymptotically. Second, the stability of the modulated DS in Euclidean space was proved in [4]. Since the modulation matrix is a rotation matrix that is full rank everywhere, it is shown that local modulation can be applied without introducing spurious attractors regardless of the original dynamics. Although only Lyapunov stability is guaranteed, we argue that in practice, unless periodic or orbital behavior is explicitly specified, the modulated system is still able to converge to its attractor if the original dynamics is GAS. In our method, the nominal DS is constructed to be GAS, and all modulation is performed in the tangent space. While an additional coordinate transformation step is applied, the underlying space remains the same tangent space; therefore, the stability guarantees of the modulated system are preserved.

V. Evaluation

In this section, we first evaluated the trajectory tracking error, that is if the modulated system is able to successfully reconstruct the modulated training data. We then proceeded to evaluate the performance of different kernels using two different metrics: accuracy and generalizability. We then concluded the evaluation with two case studies in simulation: we first showcase how modulation can be applied to optimize the manipulability in orientation space to achieve a better robot's configuration, and then show how a robot can avoid obstacle using modulation method from an orientation perspective.

A. Trajectory Tracking Error

The GPR-predicted modulation is evaluated via one-step quaternion reconstruction. For each input $q_{\text{in},i}$ in the evaluation set, the prediction $\hat{\mathbf{y}}_i$ is used to construct M_i and produce a reconstructed output $\hat{q}_{\text{out},i}$. The geodesic orientation error is:

$$e_{q,i} = \left\| \log_{q_{\text{out},i}}(\hat{q}_{\text{out},i}) \right\|_2 . \tag{17}$$

Generalization is assessed via leave-one-trajectory-out (LOTO) cross-validation across $N_{\text{IC}} = 2$ training and $N_{\text{val}} = 2$ held-out validation trajectories. The Matérn-$5/2$ kernel with $\hat{\sigma}_f^2 = 0.993$, $\hat{\ell} = 0.086$, $\hat{\sigma}_n^2 \approx 10^{-6}$ (LML $= 4952.3$) achieves $R^2 = 0.469 \pm 0.080$, the best among all kernel configurations tested (see Table I).

TABLE I: Leave-one-trajectory-out CV of GPR modulation kernel ($N = 149$ steps/traj, $N_{\text{traj}} = 2$).

Kernel	ν	ℓ	R^2 (mean $\pm$ std)	MSE
Matérn	2.5	0.50	0.469 ± 0.080	0.1345
Matérn	0.5	0.50	0.430 ± 0.137	0.1501
Matérn	1.5	0.50	0.384 ± 0.109	0.1496
RBF	–	0.50	0.276 ± 0.099	0.1832

B. Kernel Selection

The choice of covariance function $k(\cdot,\cdot)$ is critical for capturing the non-linear corrections required for manipulability adaptation. We evaluated several kernel configurations, including the Radial Basis Function (RBF) and the Matérn kernel with varying smoothness parameters $\nu \in \{0.5, 1.5, 2.5\}$. Model selection was performed via leave-one-trajectory-out (LOTO) cross-validation on a dataset of $N = 298$ points (2 demonstrations). As shown in Table I, the Matérn $\nu = 2.5$ kernel outperformed the RBF kernel, achieving a higher R^2 score (0.469 ± 0.080) and lower Mean Squared Error (MSE). The Matérn kernel's ability to model functions that are only twice-differentiable proved better suited for the transient nature of the orientation corrections compared to the infinite smoothness assumed by the RBF kernel. The final model was trained using the Matérn $\nu = 2.5$ kernel with optimized hyperparameters $\sigma_f^2 = 0.993$ and

lengthscale $\ell = 0.086$. To handle observation noise and ensure numerical stability during Cholesky decomposition, a white noise kernel with $\sigma_n^2 \approx 10^{-6}$ was added.

C. Obstacle Avoidance

In this scenario, we evaluate the framework's ability to adapt a nominal trajectory to avoid a predefined spatial obstacle. Because the original dynamical system policy is inherently rigid and unaware of the environment, executing the baseline motion would result in a collision. We noted that instead of modulating position which is a more common way for obstacle avoidance, our method opens up a new and more expressive way for obstacle avoidance in orientation space. We generated offline corrective trajectories that adjust the end-effector's orientation to steer the robot safely around the object. These offline demonstrations serve as the training data to extract our geometric modulation parameters. If a vision system is deployed to estimate the pose of the obstacle, then our method can be used for online modulation as well.

As depicted in Figure 2, the GP regression model successfully learns the mapping from the local orientation states to the required axis-angle rotation and speed-scaling parameters. During runtime execution, the modulated policy correctly anticipates the obstacle and reshapes the vector field to avoid it. Once the obstacle is cleared, the system to safely fall back to the convergence-guaranteed nominal dynamical system.

D. Manipulability Maximization

To demonstrate the flexibility of our approach beyond offline demonstrations, we applied the framework to an online optimization task focusing on kinematic manipulability, as shown in Figure 3. In this setup, an online optimization oracle continuously queries the kinematics of the 7-DoF Franka Panda robot during the pouring task.

By computing the gradient of the Yoshikawa manipulability index, $w(\boldsymbol{\theta}) = \sqrt{\det(JJ^\top)}$, with respect to the end-effector's orientation, we dynamically generate target orientations that steer the arm into better-conditioned postures away from kinematic singularities. These gradient-derived targets act as the modulation data for the GP. The model effectively learns to predict the required geometric corrections, seamlessly reshaping the nominal vector field to maximize manipulability on the fly. This confirms that our orientation modulation strategy can successfully integrate with real-time optimization metrics to dynamically satisfy execution-time objectives.

VI. Conclusion

This paper introduced a novel framework to address the inherent rigidity of standard Dynamical System (DS) motion policies by enabling the adaptive modulation of orientation trajectories. We demonstrated that necessary corrections to a nominal DS can be geometrically parameterized as a compact 4-dimensional vector, comprising a speed scaling factor and an axis-angle rotation within the attractor-aligned quaternion tangent space. To learn these state-dependent corrections from either offline demonstrations or an online oracle, our framework leverages Gaussian Process regression. Through rigorous evaluation, we found that a Matérn 2.5 kernel was particularly effective at modeling the transient, non-linear nature of these orientation corrections. Crucially our approach maintains the rigorous safety and stability certificates of the underlying DS, ensuring the system gracefully falls back to the convergence-guaranteed nominal baseline.

VII. Future

A natural next step is to transfer the learned LPV-DS and GPR modulator to a physical Franka Panda robot, where real-time Jacobian computation and hardware joint limits require torque-level control with explicit null-space damping. The GPR model could be adapted online as new demonstrations are collected, for example via incremental Gaussian process regression or variational sparse approximations, enabling the system to refine the manipulability-shaping policy during task execution without full retraining. Finally, the modulation framework could be extended to more dexterous manipulation tasks involving contact, such as cap screwing or in-hand reorientation, where coupling between position and orientation dynamics and more complex manipulability landscapes would test the scalability of the approach.

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